

THEORY OF MICROSTRUCTURAL DEVELOPMENT DURING RAPID SOLIDIFICATION

W. KURZ, B. GIOVANOLA and R. TRIVEDI†

Department of Materials Engineering, Swiss Federal Institute of Technology, Lausanne, Switzerland

(Received 23 May 1985; in revised form 23 August 1985)

Abstract—A theoretical model is presented for describing columnar (directional) growth of dendrites including growth rates in the range of the limit of absolute stability. A maximum growth rate for dendrites is predicted which is slightly below the limit of absolute stability of a planar interface. The changes in microstructures which may occur at high growth rates are discussed. Both the effect of the temperature dependant diffusion coefficient and the velocity and temperature dependant partition coefficient on microstructure characteristics are considered. The model is applied to the Ag–Cu system where detailed microstructural analysis has been published. The theoretical results are found to be in reasonable agreement with experimental data obtained from the literature.

Résumé—Un modèle théorique est présenté pour décrire la croissance colonnaire (orientée) des dendrites jusqu'à des vitesses de croissance proches de la limite de stabilité absolue. Une vitesse maximale, légèrement inférieure à la limite de stabilité absolue du front plan, est prédite pour les dendrites. Les changements de microstructures pouvant intervenir à grande vitesse sont discutés. La dépendance du coefficient de diffusion vis-à-vis de la température, celle du coefficient de partage vis-à-vis de la vitesse et de la température, et leur effet sur les paramètres microstructuraux sont considérés. Le modèle est appliqué au système Ag–Cu, dont une analyse microstructurale détaillée a été publiée. Les résultats théoriques sont en accord raisonnable avec les valeurs expérimentales tirées de la littérature.

Zusammenfassung—Ein theoretisches Modell zur Beschreibung des gerichteten Dendritenwachstums bis zu Geschwindigkeiten der absoluten Stabilität wird vorgelegt. Es wird eine maximale Wachstumsgeschwindigkeit vorausgesagt, die etwas unter der Grenzgeschwindigkeit der absoluten Stabilität einer ebenen Grenzfläche liegt. Die Gefüge bei hohen Erstarrungsgeschwindigkeiten werden diskutiert. Hierbei wird die Wirkung des temperaturabhängigen Diffusionskoeffizienten sowie des geschwindigkeits- und temperaturabhängigen Verteilungskoeffizienten auf das Gefüge berücksichtigt. Das Modell wird mit Messungen am Ag–Cu System verglichen. Eine zufriedenstellende Übereinstimmung zwischen den theoretischen Ergebnissen und den experimentellen Beobachtungen aus der Litteratur wird gefunden.

INTRODUCTION

Processes involving rapid solidification of alloys have a great practical potential [1]. Refined microstructures, extended solid solubilities, microsegregation-free crystals and new metastable phases obtained at high cooling rates may lead to an increase in fatigue, wear, or corrosion resistance of materials. The conditions for obtaining microsegregation-free structures are of specific interest as they are connected with a loss of solidification microstructure: at high growth rates, dendrites, cells or eutectic structures cannot form any more, and a homogeneous solid is produced.

Two different mechanisms can cause microsegregation-free solidification: steady-state planar growth with equilibrium partitioning at low growth rates and positive temperature gradients, and solidification at growth rates above the limit of absolute stability. The principle of constitutional supercooling was the first

theoretical support to predict conditions of planar growth with equilibrium partitioning [2]. This theory, however, does not take into account the effect of surface tension and fails therefore to explain the stabilization of the interface at high velocities [3–5]. This effect, known as the absolute stability, has been correctly predicted by Mullins and Sekerka [3], and applied to rapid solidification conditions by Coriell and Sekerka [4]. At the high rates of absolute stability, one cannot assume equilibrium partitioning any more. Such departure from equilibrium partitioning was discussed by Boettinger *et al.* [6] who used two response functions to describe the temperature and the solid composition at the interface, where the variation of the partition coefficient with growth velocity was given by Aziz [7] and Jackson *et al.* [8].

The transition from cells or dendrites to microsegregation-free structures was experimentally determined in detail by Boettinger *et al.* [9], who performed electron beam melting and solidification of Ag–Cu alloys. For alloys with small solute contents, the observed transition was in very good agreement with the theory of absolute stability. For copper

†Present address: Ames Laboratory, USDOE and Department of Materials Science and Engineering, Iowa State University, Ames, IA 50011, U.S.A.

concentrations above the maximum retrograde solubility, a banded structure was observed over a range of growth rates between dendrites and microsegregation-free structures. Such banded structures have been observed before by Beck *et al.* [10].

In electron beam or laser surface treatment, the liquid solidifies directionally from its own substrate, so there is no nucleation barrier. Only growth phenomena produce the microstructures. If the theory of directional dendrite growth at low and intermediate growth rates is well established [11–15], no model has been proposed yet to describe the microstructural development in the range of velocities close to absolute stability.

The object of this paper is to present a theoretical model for directional solidification which is valid under high growth rate conditions. In this model, an appropriate stability criterion is coupled with the solute trapping effect to predict microstructural features as a function of velocity. Furthermore, at high growth rates, the undercooling can become very large so that the variation in the diffusion coefficient with temperature must also be taken into account. In addition, changes in phase diagram characteristics with temperature should also be examined. In this model, the effects of all these variables on interface curvature and on solute composition in the solid have been considered, and their influence on microstructural characterization have been evaluated quantitatively. The model is applied to the Ag–Cu system for which detailed experimental microstructural analysis is available in the literature [9]. A reasonable agreement between the theory and the experimental data is found.

THE MODEL

Langer and Müller-Krumbhaar [16] have shown that dendrite tips do grow closely to a state which can be called marginally stable. Even if the details of this behaviour are still unclear [17], the concept has proven to be extremely useful in conciliating theory with experiment [11–15, 18]. We can therefore write the approximate relationship

$$R = \lambda_s \quad (1)$$

where R is the tip radius of a dendrite growing at steady state and λ_s is the critical wavelength of the solid–liquid interface at the limit of stability. As the instability wavelength of a sphere and of a planar interface differ only by a numerical factor which is close to unity [18], λ_s of the planar interface will be used for simplicity in the present model. After Mullins and Sekerka [3], the wavelength of the marginally stable plane front is given by

$$\omega^2 \Gamma = m G_c \xi_c - G \quad (2)$$

with $\omega = 2\pi/\lambda_s$ and

$$\xi_c = 1 - \frac{2k}{\{1 + [4\pi D/(\lambda_s V)]^2\}^{1/2} - 1 + 2k}$$

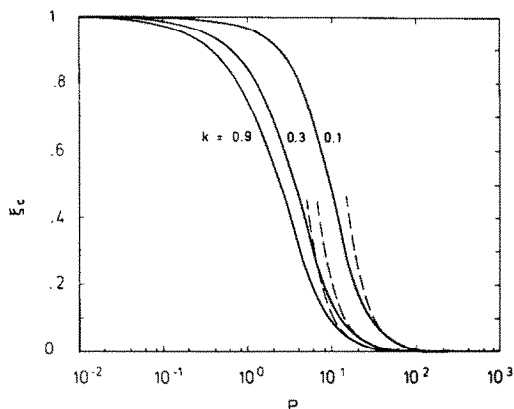


Fig. 1. Function ξ_c from equation (3). The dashed lines are the approximation $\pi^2/(kP^2)$ at high P .

Γ is the Gibbs–Thomson parameter, defined as the ratio of the specific liquid–solid interface energy to the melting entropy, m is the liquidus slope, G_c is the concentration gradient in the liquid at the interface, G is the mean temperature gradient at the interface, k is the partition coefficient, D is the liquid inter-diffusion coefficient and V is the interface velocity.

Equation (2) has been derived for the limit of small thermal Peclet numbers. Note that in metals the thermal Peclet number is very much smaller than the solutal Peclet number by the factor D/a ($\sim 10^{-2} - 10^{-4}$), where a is the thermal diffusivity.

Using equation (1), and by defining the solutal Peclet number $P = RV/2D$, one obtains

$$\xi_c = 1 - \frac{2k}{[1 + (2\pi/P)^2]^{1/2} - 1 + 2k} \quad (3)$$

and from equation (2)

$$R = 2\pi \left[\frac{\Gamma}{m G_c \xi_c - G} \right]^{1/2}. \quad (4)$$

Equation (3) is shown in Fig. 1. At high Peclet numbers ($P > \pi^2/\sqrt{k}$), it reduces to

$$\xi_c = \frac{\pi^2}{kP^2} \quad (5)$$

and substituting this limiting value in equation (2), with $G = 0$, gives the absolute stability condition.

In order to solve equation (4), G_c and G must be known. In directional growth, the temperature gradient G is determined by the external heat flow. Neglecting its effect (at low growth rates) on the diffusion field around the tip, the concentration gradient G_c can be calculated using a flux balance at the dendrite tip. Following the approach by Kurz and Fisher [12], but using Ivantsov's solution for the transport problem [19], one obtains

$$V^2 A + V B + C = 0 \quad (6)$$

with

$$A = \frac{\pi^2 \Gamma}{P^2 D^2}$$

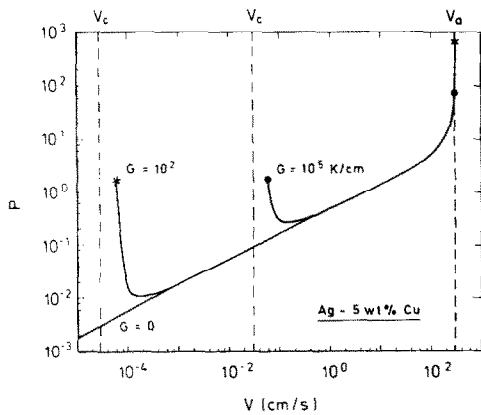


Fig. 2. Solute Peclet number P as a function of growth rate V and temperature gradient G for a Ag-5 wt%Cu alloy. V_c defines the low V limit for a planar interface, and V_a the limit of absolute stability, both according to Mullins and Sekerka. D , k_0 and m are assumed to be constant (Table 1).

$$B = \frac{mC_0(1-k)\xi_c}{D[1-(1-k)Iv(P)]}$$
$$C = G.$$

C_0 is the initial liquid composition and $Iv(P) = P \exp(P)E_1(P)$ is Ivantsov's solution, where E_1 is the exponential integral [20]. It must be noted that no special assumptions have been made about m and k . For the moment, we shall assume that these are constants.

RESULTS

Equation (6) is solved numerically to obtain the variation in solute Peclet number with velocity for different temperature gradient conditions, and the results are shown in Fig. 2. These calculations were carried out for Ag-rich Ag-Cu alloys, and the physical properties of these alloys are given in Table 1. Figure 3 shows the variation in the dendrite tip radius as a function of velocity for different compositions.

A sharp increase in P and R is observed at low as well as high velocity values. For small temperature gradients, P and R approach infinity as V approaches V_a , where V_a is the absolute velocity predicted by Mullins and Sekerka [3]. At high velocities, the temperature gradient has only a minor effect on the dendrite tip radius. However, for high gradient condi-

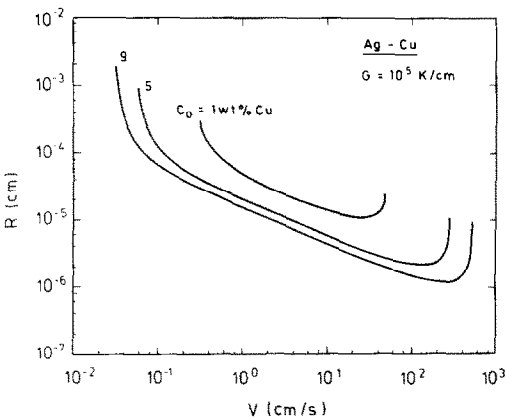


Fig. 3. Dendrite tip radius R as a function of growth rate V and initial liquid composition, with constant D , k_0 and m for several Ag-Cu alloys at $G = 10^5$ K/cm.

tions, the stability condition imposes a limiting value for the radius, and correspondingly a limiting velocity (V_{max}), above which a parabolic dendrite cannot grow with a stable tip radius. This maximum radius is found to decrease slightly as the temperature gradient is increased, although the general shape of the P - V or R - V curve is noted affected (Fig. 2). Note that V_{max} and V_a are calculated using the stability model with the supersaturation $\Omega = Iv(P)$ and $\Omega = 1$, respectively. Hence, the limit V_{max} is strictly not equal to V_a , except when $G = 0$. However, the difference is insignificant, even at large temperature gradients, because $Iv(P)$ approaches unity at high velocities. Further, from a microstructural viewpoint, the transition from dendrite to cell appears to occur before the velocity reaches V_{max} , i.e. near the velocity where a sharp increase in radius occurs.

The variation in the dendrite tip temperature T_i with velocity is calculated using the procedure described in the Appendix. Figure 4 shows the T_i - V curve for constant values of D , k_0 and m (see Table 1). The tip undercooling due to the curvature effect is also plotted, and it is seen that the largest curvature effect corresponds to the minimum radius value, as shown in Fig. 3. Note that, although the curvature effect becomes quite small near V_a , the diffusion distance also becomes very small, so that these two effects are of the same magnitude near V_a [5].

Table 1. Physical properties of the Ag-Cu system

T_f	1233.5	K	(for $T = T_f$)
Γ	$1.53 \cdot 10^{-5}$	K · cm	
k_0^a	0.31	—	
$\bar{m}^a$	-6.46	K/wt%	
$\bar{D}$	2.0×10^{-5}	cm ² /s	
D_0	1.22×10^{-3}	cm ² /s	
Q	10^4	cal/mol	
a_0	50	Å	

^aThe average values are determined by linear interpolation between the temperature of fusion of the pure component and the eutectic point.

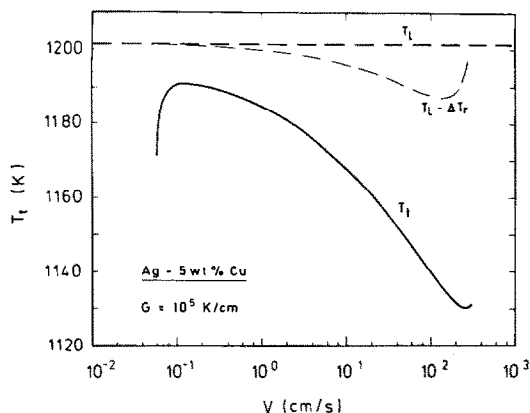


Fig. 4. Tip temperature T_t as a function of growth rate V for a Ag-5wt%Cu alloy with constant $\bar{D}$, k_0 and $\bar{m}$ and $G = 10^5$ K/cm.

Temperature dependant diffusion coefficient

As pointed out by Boettinger *et al.* [6, 9], the usual assumption of a constant liquid diffusion coefficient is invalid for rapid solidification, especially for concentrated alloys as the tip undercooling might become quite large in this case. The dependance of D on temperature can be easily introduced into the model [equation (6)], using the well known relationship

$$D = D_0 \exp[-Q/R_g T] \quad (7)$$

where D_0 is a proportionality constant, Q is the activation energy (see Table 1) and R_g is the gas constant. An iterative method is used to calculate the tip temperature.

Velocity dependant partition coefficient

Several models have been proposed to describe the dependance of the partition coefficient on velocity [7, 8, 21]. These have been discussed by Boettinger *et al.* [6]. Here, the model formulated by Aziz [7] and Jackson *et al.* [8] is used, which is

$$k = \frac{k_0 + (a_0 V/D)}{1 + (a_0 V/D)} \quad (8)$$

where k_0 is the equilibrium partition coefficient, and a_0 is a length scale related to the interatomic distance and is estimated to be between 0.5 and 5 nm, giving for D/a_0 400 and 40 cm/s respectively.

The effect of both temperature dependant diffusion coefficient and velocity dependant partition coefficient on the V - R relationship is demonstrated in Fig. 5. For each case the corresponding limit of absolute stability, V_a , for a planar front, according to Mullins and Sekerka, was also calculated and plotted in the figure.

Phase diagram effect

In solidification theory, the equilibrium partition coefficient as well as the liquidus slope is often assumed constant. This is valid only for small undercoolings, as can be seen from Fig. 6. The Ag-Cu

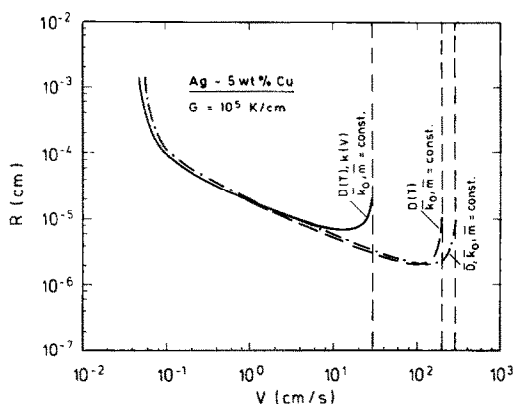


Fig. 5. Effect of the temperature dependant diffusion coefficient $D(T)$ and velocity dependant partition coefficient $k(V)$ on the R - V relationship for a Ag-5wt%Cu alloy at $G = 10^5$ K/cm.

system is a typical example of a phase diagram with a retrograde solidus. In such a system a planar front cannot grow in steady state with initial liquid compositions above the maximum retrograde solubility. However, dendrite growth can still be calculated with the present model using the function $g(C_L^*)$ in equation (A4) which can be obtained by fitting the liquidus line by a polynomial. The slope m is then calculated by simple derivation. Values of k_0 at different T are measured on the phase diagram [9] and the functional relationship $k_0(T)$ is also approximated by a polynomial (Fig. 7). The coefficients of the polynomials used are given in Table 1.

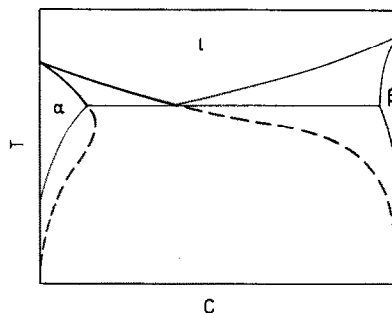


Fig. 6. Ag-Cu phase diagram with retrograde metastable solidus ($k_0 \rightarrow 0$ when ΔT increases).

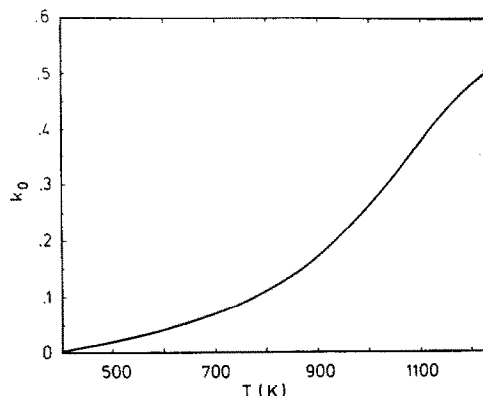


Fig. 7. Temperature dependence of the equilibrium partition coefficient k_0 for Ag-rich Ag-Cu alloys.

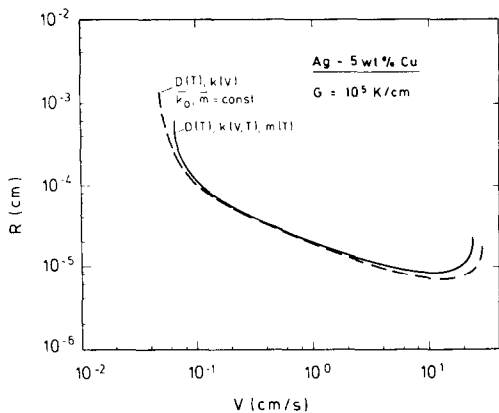


Fig. 8. Effect of the phase diagram $k(T)$, $m(T)$ on the R - V relationship for a Ag-5wt%Cu alloy at $G = 10^5$ K/cm.

The effect of the phase diagram on the R - V relationship is shown in Fig. 8. The effects of the temperature dependant diffusion coefficient, velocity dependant k , and the phase diagram on the variation in solute composition in solid, C_s^* , with velocity are shown in Fig. 9.

DISCUSSION

The model presented in this paper shows significant changes in the dendrite tip radius as the velocity approaches the absolute stability value for planar interface growth. A consistent behavior is predicted in that the radius goes to infinity and the tip temperature goes to the solidus temperature as the velocity approaches V_a . Although a small effect of temperature gradient is observed, which is due to the assumption of a parabolic interface, the magnitudes of V_{max} and V_a are nearly identical.

In order to examine the dendrite characteristics at high velocities (or high Peclet numbers), one should compare the results of the model with the theoretical predictions of Langer and Müller-Krumbhaar which have been derived for low Peclet number conditions.

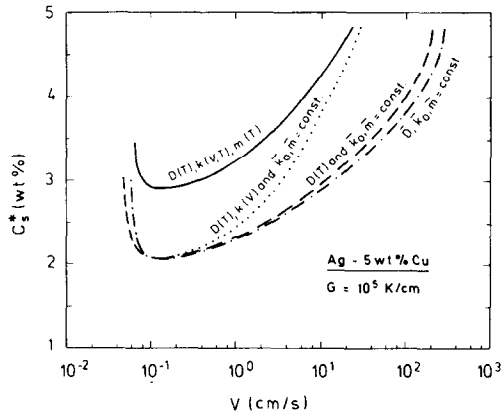


Fig. 9. Effect of the temperature dependant diffusion coefficient, $D(T)$, velocity dependant partition coefficient, $k(V)$, and phase diagram, $k(T)$, $m(T)$, on the solute composition in solid C_s^* at the dendrite tip for a Ag-5wt%Cu alloy at $G = 10^5$ K/cm.

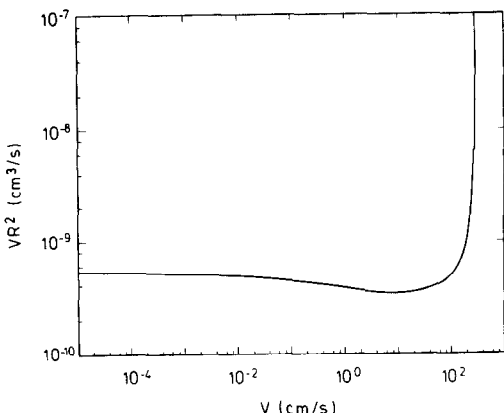


Fig. 10. VR^2 from equation (9) as a function of velocity for a Ag-5wt%Cu alloy with zero temperature gradient. The value at low velocity corresponds to $4\pi^2 D\Gamma / (k\Delta T_0)$.

Figure 10 plots the variation of the function VR^2 with velocity. A constant value is observed at small velocities ($G = 0$) where the Peclet number is small, and the current model coincides with the L-MK model. However, at high velocities, VR^2 increases sharply. Note that a minimum is observed in VR^2 and this can be understood by examining equation (6) for $G = 0$, which gives

$$VR^2 = \frac{4\pi^2 D\Gamma [1 - (1 - k)Iv(P)]}{C_0 m (k - 1) \xi_c} \tag{9}$$

For small P , $\xi_c \rightarrow 1$ and $Iv(P) \rightarrow 0$. Therefore

$$VR^2 = 4\pi^2 D\Gamma / k\Delta T_0 = \text{constant}, \quad (P \ll 1) \tag{10}$$

where $\Delta T_0 = mC_0(k - 1)/k$.

At high P , if ξ_c is assumed to be unity, then

$$VR^2 = 4\pi^2 D\Gamma / \Delta T_0. \tag{11}$$

Equation (11) differs from equation (10) by a factor $k(<1)$, which shows that VR^2 should decrease as the Peclet number (or velocity) is increased. However, if the full stability function at high velocity, ξ_c , is considered [equation (3)], then VR^2 goes to infinity. Thus, as the velocity is increased, VR^2 initially decreases, but then it increases sharply as the effect of ξ_c becomes important.

Such a minimum is not found in the pure thermal case [5] (there is no temperature discontinuity at the dendrite tip), so that VR^2 stays constant when ξ_c is assumed to be unity, and increases monotonally when the real ξ_c is considered.

Microstructural transitions

Figure 11 summarizes for a wide range of concentrations in the Ag-rich Ag-Cu system the different situations discussed before and their influence on the maximum limit of dendrite growth. Two major effects can be observed. First, the temperature dependance of the diffusion coefficient introduces a maximum in the V_{max} - C_0 relationship: when the concentration increases, the tip temperature is lowered, due to higher solute built up which in turn decreases the

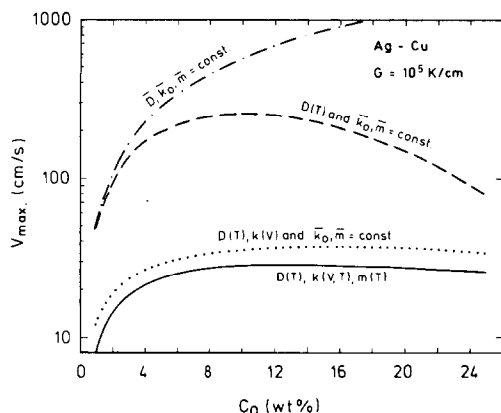


Fig. 11. Maximum rate of dendrite growth $V_{\max}$ as a function of initial liquid composition. The maximum in the interrupted curve is due to the temperature dependant diffusion coefficient $D(T)$. Solute trapping, expressed by the growth rate dependant partition coefficient $k(V)$ (points), strongly reduces $V_{\max}$. The continuous line results when the temperature dependence of k and m is taken into account.

diffusion coefficient. Secondly, the velocity dependence of the partition coefficient k strongly decreases the maximum dendrite growth rate. The influence of the real shape of the phase diagram on $V_{\max}$ (full line in Fig. 11) is small as solute trapping considerably reduces the undercooling.

Until now, $V_{\max}$ has been referred to as the limit of dendrite growth. However, a dendrite-to-cell transition has also been observed at high velocities [9]. At low growth rates, this transition appears near the minimum in the P - V relationship, that is when P begins to increase strongly [14, 15]. In the absence of a physical basis to define this transition at high growth rates, it will be assumed to occur, in analogy, at a rate where P increases strongly, that is near the minimum in the R - V relationship. The corresponding velocities, noted $V_{(R_{\min})}$, are plotted in Fig. 12 as a function of C_0 , together with the $V_{\max}$ - C_0 curve calculated with the model (continuous line in Fig. 11).

Up to the maximum retrograde solubility (~ 10 wt% for the Ag-Cu system), a planar front is encountered at velocities above $V_{\max}$. As mentioned before, the planar interface cannot grow at higher liquid compositions under steady state conditions, unless the solute trapping causes $C_s^* = C_0$. In general, it is still unclear what structures form in this case when the critical velocity $V_{\max}$ is reached. In their electron beam experiments on Ag-Cu alloys, Boettinger *et al.* [9] observed a transition from dendrites or cells to microsegregation-free structures for 1 wt% and 5 wt% copper. From 9 wt% and up to the eutectic composition, experiments revealed a structure consisting of bands formed parallel to the local solidification interface.

†Boettinger *et al.* [9] presented their results in terms of scan velocities. The corresponding growth velocity can be deduced by multiplying the scan velocity by the cosine of the angle between the scan direction and the local growth direction, as given by the same authors.

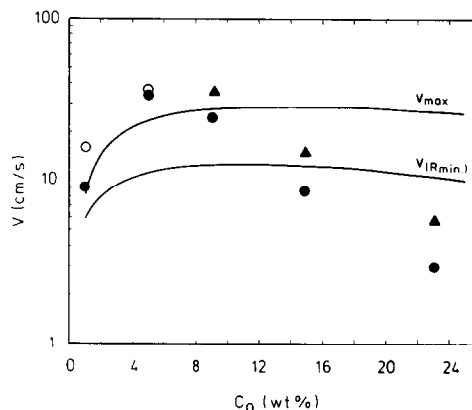


Fig. 12. Maximum rate of dendrite (or cell) growth, $V_{\max}$, as a function of initial liquid composition for Ag-Cu alloys. $V_{(R_{\min})}$ is also plotted as a possible transition from dendrites to cells. The circles and triangles are experimental data at the transition point from Boettinger *et al.* [9]. Open circles denote microsegregation-free structures, solid circles denote cells or dendrites and triangles denote banded structures.

The maximum velocities at which Boettinger *et al.* still observed cells or dendrites are also reported on Fig. 12. In the region of concentrations where a planar front can grow, the experimental points are in very good agreement with the model. Furthermore, there seems to be experimental support for the maximum in the $V_{\max}$ - C_0 predicted by the theory. The correspondence is not so good at higher concentrations: even the lower $V_{(R_{\min})}$ limit is a factor two to three higher than the experimental results at concentrations near the eutectic. These results can nevertheless be considered as satisfactory, especially in view of the uncertainty in the velocities measured† and the lack of understanding about the exact nature of the banded structure.

Solute trapping

Both Figs 11 and 12 show that the solute trapping influences quite appreciably the velocities at which the dendrite-to-cell transition occurs, and the conditions under which microsegregation free growth occurs. It would thus be desirable to propose critical experiments through which the solute trapping effect can be checked quantitatively.

One of the experimental parameters that can be measured quite accurately is the solute content in solid along the dendrite or cell axis. Figure 9 shows the effect of $D(T)$, $k(V, T)$ and $m(T)$ on the solute content. Since these effects alter the composition appreciably, we need to precisely calculate the effects of $D(T)$, $k_0(T)$ and $m(T)$ on solute composition. Once this is done only one unknown exists, i.e. a_0 in equation (8). Figure 13(a) shows the predicted variation in solute content as a function of a_0 for a Ag-5 wt%Cu alloy. Note that at high velocities, significant differences in composition exist for different a_0 values.

Hence, one should be able to determine a_0 experimentally by using Fig. 13(a), assuming that the

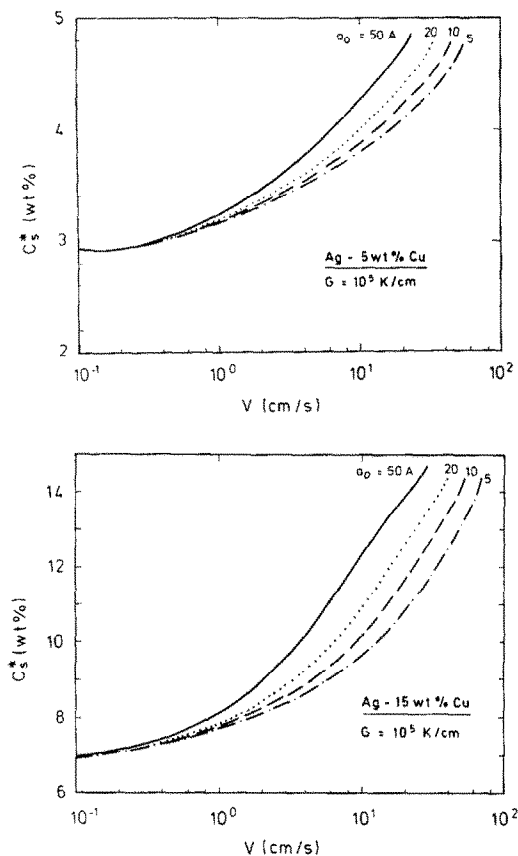


Fig. 13. Solute content in solid at the dendrite tip as a function of growth rate for different values of a_0 . (a) For a Ag-5wt%Cu alloy. (b) For a Ag-15wt%Cu alloy.

composition at the center of a dendrite or cell after complete solidification is close to the concentration at the tip during growth (negligible back diffusion). Furthermore, the theory of solute trapping does not take into account the composition dependence of a_0 . Consequently, it is desirable to compare experimental solute concentrations for other alloys to check the validity of this assumption. For example, Fig. 13(b) shows the effect of a_0 on solute concentration in the solid for the Ag-15 wt%Cu alloy.

CONCLUSION

Using the stability analysis from Mullins and Sekerka, the criterion of marginal stability from Langer and Müller-Krumbhaar has been applied to Ivantsov's solution for the mass transport at the dendrite tip. The model presented describes columnar growth of dendrites up to growth rates of absolute stability. At low velocities (small Peclet numbers), the model approaches Langer and Müller-Krumbhaar's result, $VR^2 = \text{constant}$, while at high velocities, VR^2 first decreases and finally goes to infinity when reaching the limit of absolute stability. Dendrites are shown to grow up to a maximum velocity which is very close to the absolute stability from Mullins and

Sekerka. The calculations clearly show that the temperature dependence of the diffusion coefficient, the velocity dependence of the partition coefficient, and the exact shape of the (metastable) phase diagram must be taken into account. Finally, it has been shown that careful measurement of the composition of the dendrite/cell center as a measure for the tip concentration should make it possible to obtain information on the parameter a_0 which controls the velocity dependence of the distribution coefficient.

The major predictions of the model are the sharp increase in tip radius near the absolute stability limit and the possible transition from dendritic to cellular structure at high velocity. Although limited experimental observations available in the literature agree qualitatively with the predictions of the model, detailed quantitative experimental studies are required to test the theory.

Acknowledgements—The authors are grateful to Dr W. J. Boettinger for stimulating discussions, and to the "Kommission zur Förderung wissenschaftlicher Forschung", Bern, and the Ames Laboratory USDOE for their financial support. Ames Laboratory is operated for the USDOE by Iowa State University, under contract No. W.7405-Eng-82, supported by the Director of Energy Research, Office of Basic Energy Sciences.

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APPENDIX

The supersaturation Ω is defined by

$$\Omega = \frac{C_l^* - C_0}{C_l^*(1 - k)}$$

(A1)

where C_l^* is the liquid concentration at the dendrite tip.
Using Ivantsov's solution for the transport problem around a paraboloid of revolution, one may write

$$\Omega = \text{Iv}(P).$$

(A2)

C_l^* can be calculated using equations (A1) and (A2)

$$C_l^* = \frac{C_0}{1 - (1 - k) \text{Iv}(P)}.$$

(A3)

The tip temperature T_i is given by

$$T_i = g(C_l^*) - \Delta T_r,$$

(A4)

where ΔT_r is the curvature undercooling, related to the tip radius by

$$\Delta T_r = \frac{2\Gamma}{R}$$

(A5)

and $g(C_l^*)$ is a function defining the liquidus line of the phase diagram (see Table I).

In the special case where m and k are constants (i.e. liquidus and solidus are straight lines), one has

$$g(C_l^*) = T_f + mC_l^*$$

(A6)

where T_f is the temperature of fusion of the pure component, so that the tip temperature becomes

$$T_i = T_f + mC_l^* - \frac{2\Gamma}{R}.$$

(A7)